

(allow relative speed of approach does not equal to relative speed of separation)

- 3 (a) The moment of a force about a point is the product of the force and the perpendicular distance from that point to the line of action of the force. **B1**

- (b) (i) Taking moments about point A and since the sum of the clockwise moments must equal the sum of the anticlockwise moments,

$$8.0(9.81)\left(\frac{7.5}{2}\right)\cos 37^{\circ} + 12.0(9.81)(7.5\cos 37^{\circ}) = T(3.8) \quad \text{C1}$$

A1

Hence $T = 2.5 \times 10^2 \text{ N}$ (to 2 s.f.)

- (ii) Since net horizontal force must be zero, $F_H = 2.5 \times 10^2 \text{ N}$
Since net vertical force must be zero,

$$F_V = 8.0 \times 9.81 + 12.0 \times 9.81 = 2.0 \times 10^2 \text{ N}$$

$$F = 3.2 \times 10^2 \text{ N} \quad \text{A1}$$

$$\tan \theta = F_V / F_H \quad \text{C1}$$

$$\square = 39^{\circ}, \text{ anticlockwise above horizontal.} \quad \text{A1}$$